

Precision endodontics with nanorobots: Targeted antimicrobial and regenerative strategies

INTRODUCTION

Predictable clinical success in endodontics remains constrained by anatomical complexity, entrenched polymicrobial biofilms, microbial persistence within dentinal tubules, and the limited reach of conventional chemomechanical strategies. Root canal disinfection remains a significant challenge, as conventional instrumentation and irrigants such as sodium hypochlorite and chlorhexidine often fail to eliminate biofilms within complex canal anatomies and have potential cytotoxicity. Persistent microorganisms are a leading cause of treatment failure, highlighting the need for precision-targeted therapies.^[1,2]

Recent progress in nanotechnology and microrobotics has catalyzed interest in smart nanorobots-engineered micro-to nanoscale systems capable of controlled navigation, site-specific therapeutic delivery, and active biofilm disruption. Precision endodontics, which integrates nano-enabled and data-driven technologies, offers a transformative approach to achieve both effective microbial eradication and biologically guided tissue preservation.^[3]

CONCEPTUAL FRAMEWORK AND RATIONALE

In endodontics, the proposed role of nanotechnology encompasses targeted delivery of antimicrobial agents, enzymes, nanozymes, or regenerative biomolecules directly to infected or inflamed regions of the root canal system. Nanorobotic devices may be actively driven or magnetically steered, in contrast to conventional irrigants, whose effectiveness is controlled by diffusion, fluid exchange, and canal shape.

This ability enables penetration into dentinal tubules, isthmuses, lateral canals, and apical ramifications that routinely elude conventional disinfection strategies.

Engineered nanoparticles such as silver (AgNPs), zinc oxide (ZnO), chitosan, bioactive glass, and polymer-based carriers have demonstrated enhanced antimicrobial efficacy by penetrating deeper into dentinal tubules and disrupting biofilms more effectively than conventional agents. Biodegradable nanoparticles such as chitosan and polylactic-co-glycolic acid also facilitate controlled drug delivery, potentially enhancing biocompatibility and sustained antimicrobial effects. These nanoagents can be integrated into irrigation solutions, intracanal

medicaments, sealers, or scaffolds to improve both disinfection and material performance.^[4,5]

NANOBOTS

Beyond passive nanoparticles, nanorobotic technology has emerged as a potential strategy for addressing persistent microorganisms in endodontics. Magnetically actuated nanobots, helical magnetic structures composed of silicon dioxide with embedded iron domains, are guided by external fields to navigate the low-Reynolds number environment of root canal systems, reaching depths and complexities beyond what passive diffusion can achieve. These nanobots have been shown to move within dentinal tubules under rotating magnetic fields and can generate localized hyperthermia or deliver therapeutic payloads to eradicate resistant pathogens.^[6,7] Root canal biofilms exhibit complex extracellular matrices, spatial heterogeneity, and adaptive resistance to commonly used irrigants and intracanal medicaments. By mechanically disrupting the biofilm architecture and releasing antimicrobial drugs locally and continuously, smart nanorobots can have a dual therapeutic effect. This synergistic interaction improves bacterial eradication when limiting periapical tissues' exposure to harmful chemicals.

RELEVANCE TO REGENERATIVE ENDODONTICS

Importantly, precision endodontics encompasses regenerative therapy and vital pulp preservation.^[8] Nanoparticle-based carriers can deliver growth factors, bioactive molecules, and stem cell-supportive agents directly to pulp or periapical tissues, promoting angiogenesis, odontoblastic differentiation, and reparative dentin formation within engineered scaffolds. These scaffold systems enhance the microenvironment for tissue regeneration, addressing not only microbial control but also biological restoration of pulp dentin complexes.

CHALLENGES

Despite their conceptual appeal, the clinical translation of nanorobotic systems is constrained by substantial scientific and regulatory challenges. Biocompatibility and long-term biosafety remain paramount concerns, particularly with respect to nanoparticle persistence, degradation products, and host tissue responses. Achieving precise *in vivo* control, real-time localization, and either retrieval or predictable biodegradation of nanorobots represents an additional

layer of technical complexity. Practical and feasibility challenges include scalable manufacturing, sterilization assurance, cost-effectiveness, and integration into existing clinical workflows. Nanorobots occupy an ambiguous regulatory space between devices, drugs, and biologics, requiring stringent preclinical validation and clearly defined risk–benefit profiles for approval.

CONCLUSION

Smart nanorobots are a promising advancement in endodontics, potentially enhancing precision in intracanal disinfection and biological modulation. Although their clinical application is still in the aspirational phase, experimental results indicate that they could significantly overcome limitations of traditional root canal therapy. The technology's effectiveness will hinge on further research, safety validation, and ethical clinical integration. While challenges remain, including biocompatibility, cytotoxicity, long-term safety, clinical translation, cost, and regulatory pathways, ongoing research points toward a future, in which precision endodontics transforms clinical practice. By combining nanotechnology, robotics, and smart drug delivery, clinicians can achieve targeted disinfection and biologically guided regeneration with unprecedented precision and predictability.

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